

MT12 Racing Lua

Pit Manual

Race dash · Drive aids · On-track reference

Version v10.0.0

RadioMaster MT12 · EdgeTX v2.12.2 "Queen Anne's Revenge"

Every screen · Every setting · Pit notes for the track

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This is your on-track reference for MT12 Racing Lua on the RadioMaster MT12 surface radio (EdgeTX v2.12.2 "Queen Anne's Revenge"). It covers every screen, every setting, and how the drive aids behave when you drive.

No install steps here — this book assumes the script is already on the radio and assigned. Think of it as pit notes: what to look at, what each knob does, and how to tune without getting lost.

Pit note: Software version: v10.0.0. Developed and tested on EdgeTX v2.12.2 "Queen Anne's Revenge".

1. What this software is

MT12 Racing Lua is a race dashboard and drive-aid suite for a pistol-grip surface radio. On the home screen you get pack voltage, laps, heat timer, and link quality. Behind the scenes it can shape throttle, brake, and steering before the mixes send them to the car.

Think of EdgeTX v2.12.2 "Queen Anne's Revenge" as the radio firmware, and this script as the car layer — race home plus ABS- and traction-style aids tuned for surface RC.

What it can do (everyday words)

- ABS — pulses the brake so the wheels lock less hard
- Soft launch (Traction Control) — eases throttle so the car does not dump full power at once
- Less power when you steer (Steer Cut Throttle) — cuts throttle a bit in turns to reduce spin-out
- Corner brake help (Corner Brake) — eases brake while you steer so the rear is less likely to lock
- Steering help at speed (Steer Limit) — shrinks steering travel as you go faster
- Laps — times laps from a button or channel (e.g. transponder / switch)
- Session timer — countdown clock for practice or a race heat (haptic at zero is always on)
- Battery alarm — warns when pack cell voltage is low

Processing order

When drive aids are active, signals are processed in this fixed order:

Traction Control → Steer Cut Throttle → Corner Brake → ABS → Steer Limit

DYN Target / DYN Min Delay (hard brake, DYNAMIC mode) and Corner ABS Hz / Corner Delay (steering / corners) are independent. Changing one family does not change the other. Both build on Pulse Delay as the baseline ABS hold.

Safety: Always test at low speed first. This is a radio aid, not magic. Wrong Brake Direction, extreme settings, or untested changes can make the car stop or launch worse, not better. Start gentle. Change one thing at a time.

Pit note: Settings live on five tabs: BRK, DRV, SET, LAP, VOLT. Open them from the Race Home screen with ENTER.

2. Controls and navigation

From Race Home

Control	What it does
ENTER	Open the settings tabs (loads the full menu)
EXIT	Leave settings and return to Race Home (when not editing a value)

Inside settings

Control	What it does
Rotary PREV / NEXT (or virtual prev/next)	Move the selection up/down a row. Past the first/last row, change tab.
ENTER	Toggle edit mode on the selected setting (not used on LAP for editing)
PREV / NEXT while editing	Change the value by the setting's step
EXIT while editing	Stop editing (keep current value)
EXIT when not editing	Return to Race Home

Tab order (left to right): BRK · DRV · SET · LAP · VOLT. On the LAP tab there are no knobs — only times — so the rotary mainly changes tabs.

Pit note: Cold start shows only the Race Home shell (light on memory). Pressing ENTER loads the full settings UI. After EXIT, the menu may stay cached for a faster next open.

3. Race Home screen

This is your race dash. It does not change settings — it shows live info while you run. Press ENTER to open settings.

Top bar

What you see	Meaning
Model name	Name of the model on the radio (shortened if long)
LQ	Link quality (signal strength number, 0–100)
SNR	Signal-to-noise — how clean the link is (or “--” if not available)
TX voltage	Radio handset battery voltage
mW	Transmit power in milliwatts (or “---mW” if not available)

Left side — battery

Label	Meaning
PACK	Whole pack voltage, plus cell count when known (e.g. 2S)
CELL	Voltage per cell
BATT	Battery remaining as a percent (from chemistry and cell voltage)
MIN	Lowest pack voltage seen this session
STATE	Battery status: OFF, OK, LOW, or ALARM

PACK can invert/highlight when STATE is LOW or ALARM (ALARM blinks). STATE details are configured on the VOLT tab.

Right side — laps

Label	Meaning
CUR	Current lap time (running)
L1, L2, L3	Last three finished laps (L1 is newest)
LAP	How many laps you have completed
BEST	Best (fastest) lap this session — flashes briefly on a new best

Footer

Two modes — pick one with Show Timer on the SET tab:

Show Timer	Footer shows
ON	Session countdown timer (minutes:seconds). Blinks inverted when expired.
OFF	Status letters A T S C D — each lights when that aid is working right now.

See Chapter 12 for the letter cheat sheet, and Chapter 6 (SET) for timer / lap channels.

4. Tab BRK — Brake / ABS

How ABS pulses the brake. Start here if stops feel locked, soft, or inconsistent. On BRK (and DRV / SET) a live scope bar sits under the list — see Chapter 9.

Setting	Range / step	What it does
ABS Mode	OFF / STATIC / DYNAMIC	OFF = no ABS. STATIC = same pulse style every time. DYNAMIC = pulse style changes as you brake harder.
ABS Frequency	5–33 Hz, step 1	How fast the brake pulses. Higher = faster chatter.
Brake Floor	0–100%, step 5	Weakest brake during a pulse — how hard the brake still grabs when ABS “releases.”
Pulse Delay	0–100 ms, step 5	Baseline full-brake hold in each ABS cycle before the next chatter. Not an “arming delay” before ABS starts.
ABS Trigger	5–100%, step 5	How deep you must pull the brake before ABS starts. Higher = waits for a harder pull.
DYN Target	5–33 Hz, step 1	DYNAMIC only: target pulse speed at hard brake. Can be higher or lower than ABS Frequency.
DYN Min Delay	OFF or 5–100 ms, step 5	DYNAMIC only: shortest hold time at hard brake. Independent of Corner Delay.

Pit note: For a full ABS walkthrough, see Chapter 10. Electric and nitro/gas brakes feel different — match Brake Direction on SET and tune Trig carefully.

5. Tab DRV — Drive / stability

Soft launch, corner help, and steering at speed. These work with throttle and steering, not only brake.

Setting	Range / step	What it does
Traction Control	OFF or 0–100%, step 5	Soft launch: how slowly throttle is allowed to rise. Higher = gentler takeoff. Rise-only — partial lift is instant.
Steer Cut Throttle	OFF or 0–100%, step 5	Less power when you steer. Higher = more cut.
Corner Brake	OFF or 0–100%, step 5	Less brake in corners. Higher = more brake ease when steering.
Corner ABS Hz	OFF or 0–7 Hz, step 1	Extra ABS chatter when you lock the wheel in a corner (adds on top of base ABS Hz). 0 = off.
Corner Delay	OFF or 5–100 ms, step 5	Shortest pulse hold from steering only. Independent of DYN Min Delay.
Steer Limit	OFF or 0–100%, step 5	How much steering “shrinks” at speed. Higher = more reduction.
Steer Limit Start	0–100%, step 5	From which throttle % Steer Limit begins. Below this, full steering stays.

Pit note: Steer Cut Throttle and Corner Brake need a correct Steer Channel on SET. If those letters never light, check the channel and that Gain is not OFF.

6. Tab SET — Setup / radio choices

Timer, channels, brake side, lap button, and small display choices. Match these to how your radio and model are set up.

Setting	Range	What it does
Timer	1–45 min	Session length for the heat countdown. Default 5 min.
Timer Channel	CH1–CH8	Channel that starts and stops the timer (press/hold style on that channel).
Show Timer	ON / OFF	ON = show the timer on the home footer. OFF = show letters A T S C D instead.
Brake Direction	NEG / POS	Which side of the trigger is brake. Must match your model / ESC / servo.
Graph Direction	NOR / REV	Flips steering direction on the live graph only. Does not reverse the car.
Steer Channel	CH1–CH8	Which channel is steering (used by Steer Cut Throttle, Corner Brake, and Steer Limit).
Laptimer Channel	CH1–CH8	Channel for lap button or transponder-style pulse.
Laptimer Haptic	ON / OFF	ON = vibrate on each new finished lap.
Throttle Start Lap	ON / OFF	ON = auto-start lap timing when you apply throttle. OFF = wait for the lap channel.

Fixed behavior (not on SET)

- Timer always vibrates when it hits zero (cannot be turned off).
- Very short “double trigger” laps under 3 seconds are ignored (debounce).

7. Tab LAP — Lap view

A bigger lap list and session stats. It is not a list of knobs — it shows times.

On screen	Meaning
CUR	Current lap running now (highlighted)
L1 ... L5	Last five finished laps (L1 is newest). Deltas may show vs previous.
AVG	Average of the stored finished laps
LAPS	Total lap count
TOT	Total session time
BEST	Best lap this session

Reset laps

Hold the lap control long enough and the screen shows a progress bar, then RESET, and clears lap data for a new session (hold about 1.2 seconds after the bar starts).

Pit note: Lap timing rules: min lap 3 s; optional Throttle Start Lap from throttle; Laptimer Haptic on new laps. Configure channels on SET.

8. Tab VOLT — Battery alarm

Tell the radio what battery you run and when to warn you. At the top you also see live PACK, CELL, and STATE (same idea as home).

Setting	Range	What it does
Battery Type	LiPo / LiHV	Battery chemistry. Affects the % estimate (full voltage differs).
Cells	Auto or 1S–6S	Auto guesses from pack voltage, or lock a fixed cell count.
Volt Alarm	ON / OFF	Master switch for the low-voltage alarm.
Alarm Cell Voltage	3.20–3.90 V, step 0.05	Alarm when cell voltage drops under this value.

STATE meanings

STATE	Meaning
OFF	Alarm disabled (Volt Alarm OFF), or not evaluating.
OK	Pack present and cells above the alarm threshold (or no pack / no cells to judge).
LOW	Cell voltage is under Alarm Cell Voltage, but the hold delay has not finished yet (about 5 s).
ALARM	Held low long enough — active alarm. Home PACK / STATE highlight; radio haptic cycles.

Haptic pattern (when alarming): short active bursts with longer quiet gaps so it does not buzz continuously. Voltage has a small hysteresis so the alarm does not chatter on the edge of the threshold.

Pit note: BATT % uses chemistry: empty is treated around 3.50 V/cell; full is higher for LiHV than LiPo. Use Battery Type to match your packs for a saner percentage.

9. Live scope bar (BRK · DRV · SET)

On BRK, DRV, and SET (not LAP or VOLT), a live bar sits near the bottom of the screen. It shows what the script is doing to throttle/brake and where steering is — useful while tuning.

What you see

Element	Meaning
Solid fill from center	Processed throttle (right of center when accelerating) or brake (left or right of center depending on Brake Direction).
Solid vertical midline	Zero / center of the stick range.
Dotted vertical line (floor)	When ABS is active: brake floor level (how much brake remains on the “release” side of the pulse).
Dotted vertical line (raw)	When aids are shaping the signal: raw stick reference if it differs from the processed bar.
Solid vertical line (trig)	When ABS is enabled but not yet active: ABS Trigger depth — pull past this for ABS to start.
Small block under the bar	Steering position (left/right of center). Graph Direction only flips this tick.
Letters A T S C D	Same live aid indicators as home footer (when Show Timer is OFF on home).
Status text (right)	READY, ABS OFF, ABS ON, or ABS DYN.

Pit note: If the bar moves the “wrong way” for brake, fix Brake Direction on SET. If only the steer tick looks mirrored, use Graph Direction — that does not reverse the car.

10. ABS guide (pit notes)

This chapter is the heart of the manual. Read it once before you race with ABS on.

10.1 What ABS does on an RC car

ABS pulses the brake on and off very quickly. That can keep the wheels from locking as hard as a full grab. On loose dirt, that often means more control and sometimes shorter stops. On sticky asphalt, you may still want ABS — but the best numbers change with surface and tires.

ABS is a radio aid. It does not replace good brake setup on the car.

10.2 Electric vs nitro / gas (mindset)

	Electric	Nitro / gas
Brake feel	Often more “linear” ESC brake	Mechanical / servo brake can feel different
Brake Direction	Match ESC / mixer	Match servo direction carefully
ABS Trigger	Start mid; ESC brake % matters	Start careful; deadband may need higher trig
Tune tip	Change one ABS knob at a time	Same — and verify failsafe / throttle cut

10.3 The three modes

Mode	What it does
OFF	No ABS. Full brake is whatever you pull.
STATIC	Same pulse speed (ABS Frequency) and floor (Brake Floor) every time ABS runs.
DYNAMIC	As you brake harder, pulse speed moves toward DYN Target. Optional DYN Min Delay shortens the hold time at hard brake.

DYNAMIC tip: DYN Target can be higher or lower than ABS Frequency. Higher target = faster chatter at full pull. Lower target = slower chatter at full pull.

10.4 Each ABS knob in one sentence

Setting	One sentence
ABS Frequency	Sets how many brake pulses per second (base rate).

Setting	One sentence
Brake Floor	Sets how much brake is left when ABS “lets go” for a moment.
Pulse Delay	Sets how long each cycle keeps full brake (baseline hold) before the next chatter.
ABS Trigger	Sets how hard you must pull brake before ABS starts.
DYN Target	In DYNAMIC, the pulse speed goal at hard brake.
DYN Min Delay	In DYNAMIC, optional shortest hold time at hard brake (OFF turns this off).

10.5 How ABS works with CBC

Corner Brake lives on the DRV tab. It can ease brake while you steer, and add a little extra ABS chatter in corners.

Setting	Simple role with ABS
Corner Brake	Softens brake when you are steered (less rear lock in corners). Step 5%.
Corner ABS Hz	Adds extra pulse speed when steering (0–7 Hz; 0 = off).
Corner Delay	In corners, can pull pulse hold toward a shorter floor from steering only.

Important: DYN Min Delay (hard brake, DYNAMIC) and Corner Delay (steering / corners) are separate. Changing one does not change the other. Both build on Pulse Delay as the baseline hold.

10.6 First-time setup (low speed)

- Open settings (ENTER), go to BRK.
- Set ABS Mode to STATIC.
- Set ABS Trigger so ABS only starts on a real brake pull — not a light touch. If ABS never starts, lower ABS Trigger. Also check Brake Direction on SET.
- Pick a mid ABS Frequency — about 10–15 Hz is a gentle starting area.
- Set Brake Floor so the car still slows down. If it barely stops, raise Brake Floor. If it still locks hard, lower Brake Floor a little.
- Adjust Pulse Delay if the chatter feels too long or too snappy (hold inside the pulse cycle).
- When STATIC feels okay, try DYNAMIC. Set DYN Target a bit above or below ABS Frequency and feel hard brake. Add DYN Min Delay only if you want shorter holds at full pull.

10.7 Quick tune tips

Feel	Try this
Brakes feel too soft / car won't stop	Raise Brake Floor, or lower ABS Frequency
Chatters too hard / feels wrong	Lower ABS Frequency, or change Pulse Delay
ABS never starts	Lower ABS Trigger, or check Brake Direction on SET
Locks hard only in corners	Raise Corner Brake; try a little Corner ABS Hz
Dirt / loose	Often a bit more floor and moderate freq — stay gentle, then tune
Asphalt / high grip	Often you can run a bit more aggressive freq — still test slowly first

These are starting ideas, not lab numbers. Every car and tire is different.

11. Drive aids deep dive

11.1 Traction Control — soft launch

Traction Control is a radio-side soft launch: it limits how fast throttle increases (rise-only). It is not a gyro system (not Spektrum-style AVC) and not a throttle pulse chopper.

- Only active when accelerating and Traction Control > 0 (not OFF).
- Partial lift / going to brake is not held back by the ramp — release is free.
- Higher % = slower rise to full throttle = gentler launch on dirt or high-power cars.
- Letter T lights on the home footer / scope when Traction Control is actively holding back rise.

11.2 Steer Cut Throttle

Steer Cut Throttle reduces throttle when you apply steering. More lock (more steer) → more cut at a given Steer Cut Throttle value. Use it to calm wheelspin mid-corner or on power out. Letter S lights when the cut is active.

11.3 Corner Brake

Corner Brake eases brake when you are steered, and can add ABS chatter / shorter hold from steering (Corner ABS Hz, Corner Delay). Letter C lights when corner brake ease is active.

11.4 Steer Limit

Steer Limit shrinks steering travel as throttle goes past Steer Limit Start. Higher Steer Limit = more shrink. Letter D lights when the reduction is active. Useful for high-speed stability without dulling low-speed steering.

11.5 Pipeline reminder

Order: Traction Control → Steer Cut Throttle → Corner Brake → ABS → Steer Limit. Each stage can leave flags that light A / T / S / C / D.

12. Status letters cheat sheet

On the home footer (when Show Timer is OFF), and on the scope status strip, letters light when that aid is working right now:

Letter	Aid	Plain meaning
A	ABS	Brake pulsing is active
T	Traction Control	Soft launch is holding back throttle rise
S	Steer Cut Throttle	Power is cut because you are steering
C	Corner Brake	Corner brake ease is active
D	Steer Limit	Steering shrink at speed is active

If a letter never lights, that aid may be OFF, or the conditions for it are not met (for example, no steer for Steer Cut Throttle / Corner Brake, not enough brake for ABS, or throttle below Steer Limit Start).

13. Session timer and laps (on track)

Session timer

- Length: Timer on SET (1–45 min). Default 5.
- Start / pause / restart: use the channel set as Timer Channel (channel press behavior).
- Home footer shows remaining time when Show Timer is ON.
- When time hits zero: status expired, footer blinks, radio always vibrates.

Laps

- Lap channel (Laptimer Channel) records a finished lap when triggered (after min 3 s).
- Throttle Start Lap ON starts the running lap clock when you apply throttle; OFF waits for lap channel activity as configured.

- Laptimer Haptic ON pulses the radio on each new finished lap.
- Home shows last 3 laps; LAP tab shows last 5 plus AVG / TOT / BEST.
- Long-hold lap control on LAP tab resets the session.

14. On-track workflows

Pre-heat checklist

- Brake Direction matches how the car brakes (pull brake on the trigger and watch the scope bar).
- Steer Channel matches real steering (Steer Cut Throttle / Corner Brake / Steer Limit).
- Laptimer Channel and Timer Channel assigned to the switches/channels you will use.
- Show Timer: timer for heats, or letters for tuning aids.
- VOLT: Chem, Cells, Volt Alarm, Alarm Cell Voltage set for today's packs.
- ABS Mode OFF or gentle STATIC until the first low-speed brake test.

Practice session flow

- Race Home for pack, link, and live laps.
- Start heat timer from Timer Channel when ready.
- After a run, open LAP for AVG / BEST / full list; long-hold to RESET for the next stint.
- Tune one tab at a time (usually BRK first, then DRV).

Something feels wrong

Problem	Try
No ABS feel	ABS Mode not OFF; lower ABS Trigger; check Brake Direction; confirm brake moves the scope bar
ABS always on light touch	Raise ABS Trigger
Car won't stop with ABS	Raise Brake Floor; lower ABS Frequency; check Pulse Delay
T letter never lights	Traction Control not OFF; apply rising throttle (not just holding)
S or C never light	Steer Cut Throttle / Corner Brake not OFF; Steer Channel correct; actually steer while on power/brake
D never lights	Steer Limit not OFF; throttle past Steer Limit Start
Laps too short / double counts	Min 3 s debounce is fixed; ease the lap button technique
Pack STATE always OFF	Turn Volt Alarm ON on VOLT; ensure pack telemetry is present
BATT % looks wrong	Set Battery Type (LiPo/LiHV) and Cells (or Auto) to match the pack
Graph steer looks mirrored	Graph Direction only — does not reverse the car

15. Quick maps

Software map

Place	What it is
Race Home	Live race view — battery, laps, timer or A/T/S/C/D
ENTER	Open settings
BRK	ABS and brake pulse settings
DRV	Soft launch, corner help, steering at speed
SET	Timer, channels, directions, lap options

Place	What it is
LAP	Lap list and session stats
VOLT	Battery type and low-voltage alarm
Scope bar	Live thr/brake/steer view on BRK/DRV/SET

Knob cheat sheet (one line each)

Knob	In one line
ABS Mode	Off, fixed pulse, or speed that changes with brake depth
ABS Frequency	Pulse rate
Brake Floor	Brake left during ABS “release”
Pulse Delay	Full-brake hold inside each pulse cycle
ABS Trigger	How hard to pull before ABS starts
DYN Target	Hard-brake pulse rate goal in DYNAMIC
DYN Min Delay	Shortest hold at hard brake (DYNAMIC)
Traction Control	How slow throttle may rise
Steer Cut Throttle	How much to cut power when steered
Corner Brake	How much to ease brake when steered
Corner ABS Hz	Extra ABS chatter from steering
Corner Delay	Shortest hold from steering / corners
Steer Limit	How much to shrink steering at speed
Steer Limit Start	Throttle % where Steer Limit begins
Show Timer	Footer timer vs aid letters
Brake Direction	Which trigger side is brake
Alarm Cell Voltage	Per-cell voltage that starts the alarm path

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